

Lab 4: Response in the Time Domain

- Review
 - Laplace transform
 - First order system response
 - * Free response
 - * Step response
 - * Impulse response
 - * Characteristics:
 - Time constant τ - significance
 - * Parameter estimation
 - Time constant may be approximated by the time it takes for the response to reach the fraction $(1 - e^{-1})$ of the initial change from initial condition to steady state value.
 - Time constant may be approximated from the slope of a semilog plot of the response

$$x(t) = x_0 e^{-\frac{t}{\tau}} \quad (1)$$

$$\log[x(t)] = \log\left[x_0 e^{-\frac{t}{\tau}}\right] \quad (2)$$

$$= \log[x_0] + \log\left[e^{-\frac{t}{\tau}}\right] \quad (3)$$

$$= \log[x_0] - \frac{t}{\tau} \log[e] \quad (4)$$

$$= \log[x_0] - \frac{t}{\tau} 0.434294 \quad (5)$$

- Second order system response
 - * Undamped vs. damped
 - * Free response
 - * Step response
 - Maximum percent overshoot
 - Peak time
 - 100% rise time
 - 2% settling time
 - Delay time
 - * Impulse response
 - * Characteristics
 - Natural frequency ω_n

- Period T
- Damping ratio ζ
- Damped natural frequency ω_d
- Time constant τ
- * Parameter estimation
 - Damped natural frequency may be approximated from the period

$$\omega_d = \frac{2\pi}{T} \quad (6)$$

- Damping ratio may be approximated from the logarithmic decrement δ

$$\delta = \frac{1}{n} \ln \left(\frac{x(t)}{x(t + nT)} \right) \quad (7)$$

$$= \frac{2\pi\zeta}{1 - \zeta^2} \quad (8)$$

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 - \delta^2}} \quad (9)$$

- Free response from periodic input
 - * Period of input much larger than time constant and period of the system
 - * Step response and initial conditions
- Equipment
 - Breadboard with triple power supply
 - Sweep/function generator (link)
 - Oscilloscope (link)
 - 741 op amp (link)
 - * $V_S^+ = 6 \text{ V}$
 - * $V_S^- = -6 \text{ V}$
 - Resistors: 8.2Ω , $1\text{k}\Omega$, $5\text{k}\Omega$, $10\text{k}\Omega$
 - Capacitors: $1 \mu\text{F}$, $0.1 \mu\text{F}$, 10 nF
 - Inductor: 33.3 mH
- **For several circuits, your overall objectives are to (i) determine experimentally the response characteristics (including corresponding uncertainty analyses) and (ii) identify and explain any differences relative to the corresponding theoretically expected response.**
 - Op amp response

- * **Define a characteristic to quantify how long it takes for the output of the op amp to respond to changes in the input.**
- * 741 Op amp (follower) response
 - Input: square wave 500 mV_{pp}
- First order system response
 - * **Determine the time constant: τ**
 - * RC circuit 1
 - R=1k Ω , C=.1 μ F (measure and record resistance and capacitance values)
 - Input: square wave 500 mV_{pp} (and optional 250mV DC offset)
 - * RC circuit 2 (with numerator dynamics)
 - R=1k Ω , C=.1 μ F (measure and record resistance and capacitance values)
 - Input: square wave 500 mV_{pp}
- Second order system response
 - * **Determine applicable response characteristics from among the following: τ , ω_n , T , ζ , ω_d**
 - * RC circuit 3
 - R=1k Ω , C=.1 μ F (measure and record resistance and capacitance values)
 - Input: square wave 500 mV_{pp} (and optional 250mV DC offset)
 - * Op amp circuit 1
 - see page 707 of [2]
 - $R_1 = R_2 = R_A = 10\text{k}\Omega$, $R_B = 6\text{k}\Omega$, $C_1 = C_2 = 0.1\mu\text{F}$ (measure and record resistance and capacitance values)
 - Input: square wave 500 mV_{pp} (and optional 250mV DC offset)
 - * Op amp circuit 2
 - see http://hasler.ece.gatech.edu/Courses/ECE6414/Unit1/Discrete_01.pdf
 - $R_1 = R_2 = R_3 = 1\text{k}\Omega$, $C_4 = 1.0\mu\text{F}$, $C_5 = 10\text{nF}$ (measure and record resistance and capacitance values)
 - Input: square wave 500 mV_{pp} (and optional 250mV DC offset)
 - * LRC circuit (with numerator dynamics)
 - R=8.2 Ω , C=1 μ F, L=33.3 mH (measure and record resistance and capacitance values including the resistance of the inductor)
 - Input: square wave 500 mV_{pp}
 - (Hint, you may want to use a follower op amp to provide an input to your system.)

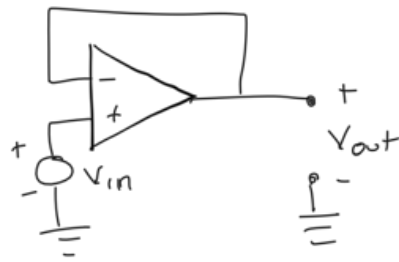
- Lab report:
 - Your lab report should follow the required lab report structure for this class
 - **Optional:** In the body of your report, you should build a strong case for your objectives to:
 1. Determine experimentally the response characteristics (including corresponding uncertainty analyses).
 - * For each circuit be sure to provide the numerical values with corresponding uncertainty for applicable response characteristics. (for example, $\tau = 5.0 \pm 0.2\text{ms}$)
 2. Identify and explain any differences relative to the corresponding theoretically expected response.
 - * For each circuit be sure to provide the theoretically expected value for each relevant response characteristic.
 - * Be sure to provide possible explanations for discrepancies, especially when the theoretically expected values are outside the experimental uncertainty.
 - Include the following in the Appendix:
 - * This lab outline (required)
 - * Data you collected during lab (required)
 - * Responses to the following questions for each circuit you built in the lab. (optional)
 - What is the theoretically expected response? Derive.
 - How did you obtain relevant response characteristics from your measurements? Provide equations.
 - How do you calculate the uncertainty of the measured response characteristics? Derive.
 - Does the response of the system differ qualitatively? Propose explanations for major discrepancies.
 - * Any other details (data, derivations, calculations, experimental procedure, screen captures etc.)

(The purpose of the Appendix in these reports is to: (i) serve as somewhat of a lab notebook, (ii) demonstrate to the grader what you did during lab, and (iii) hold material that didn't fit into the body of the report but would be necessary to repeat the experiments or justify the claims in the report.)

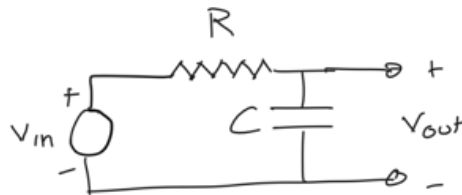
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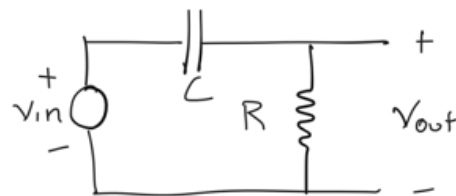
1) Follower



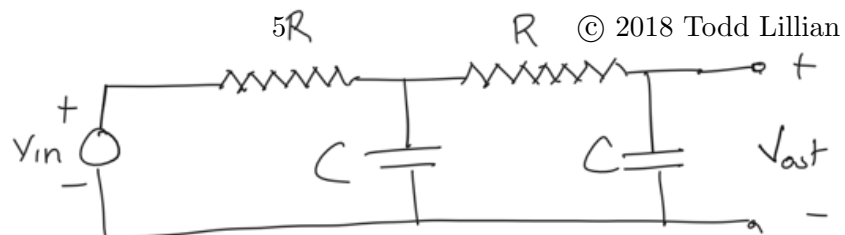
2) RC circuit 1



3) RC circuit 2

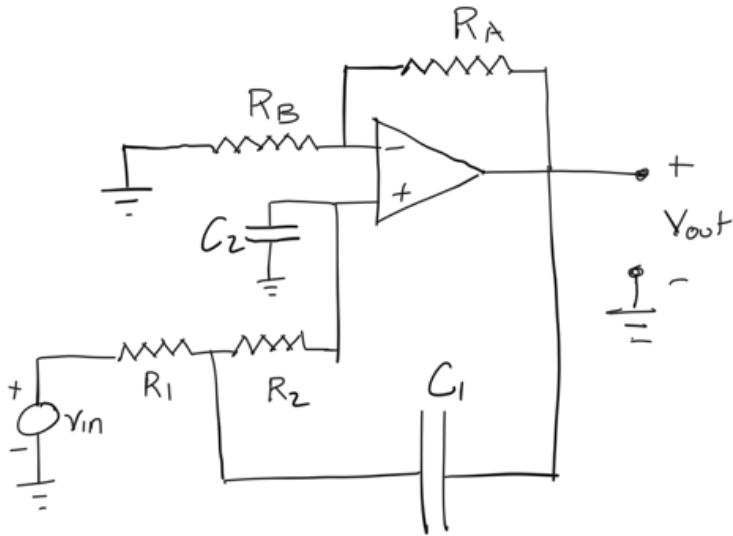


4) RC circuit 3



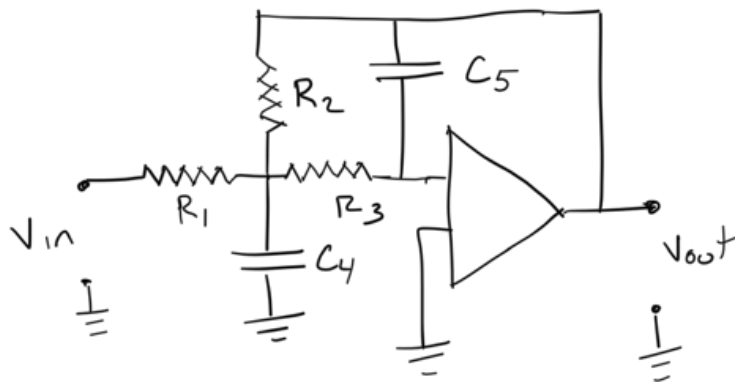
5) Op Amp Circuit 1

[See Rizzoni]



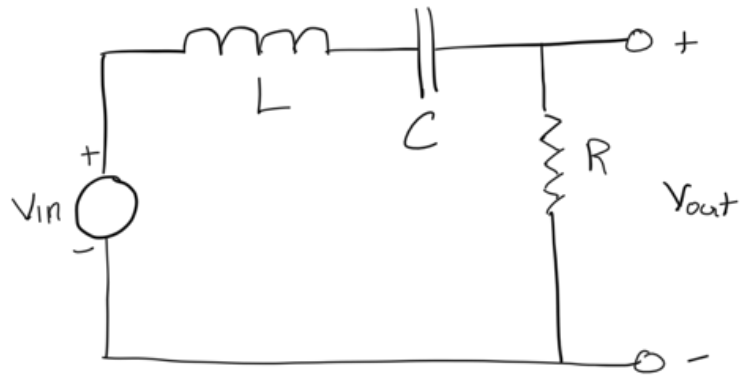
6) Op Amp circuit 2

see [hasler.ece.gatech.edu/Courses/ECE6414/Unit1/Discrete_O1.pdf]



7) LRC circuit

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References

- [1] Richard S. Figliola and Donald E. Beasley. *Theory and Design for Mechanical Measurement, 3rd Ed.* JOHN WILEY & SONS INC, 2000.
- [2] Georgio Rizzoni. *Principles and Applications of Electrical Engineering, 3rd ed.* McGraw-Hill Companies, Inc, 2000.
- [3] Anthony J. Wheeler and Ahmed R. Ganji. *Introduction to Engineering Experimentation, 3rd Ed.* Prentice Hall, 2010.